

PPE compliance system

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Sprint plan for 15 days

Sprint 1 (Day 1 – Day 3):

Goal is to setup project and backend foundation

Tasks:

- Set up development environment and GitHub repository
- Configure Supabase database and create required tables
- Implement backend configuration (config.py)
- Develop database CRUD operations (db.py)
- Configure environment variables (.env)
- Implement JWT authentication and user login

MVP:

A secure backend connected to Supabase with authentication and working database operations.

Sprint 2 (Day 4 – Day 6):

Goal is to integrate ML model and APIs

Tasks:

- Integrate YOLOv8 PPE detection model
- Develop detector.py
- Implement image upload detection API
- Create Camera, Detection, Event and Alert REST APIs
- Validate API responses using Swagger

MVP:

Backend capable of accepting images, performing PPE detection, and storing detection events in the database.

Sprint 3 (Day 7 – Day 9):

Goal is to setup live camera streaming and real-time detection

Tasks:

- Develop camera_manager.py
- Implement stream.py
- Integrate IP Webcam/CCTV feed
- Perform real-time YOLO inference
- Generate bounding boxes on live frames
- Automatically classify compliance, violations, and falls

MVP:

Live camera feed connected to the backend with real-time PPE detection and annotated video stream.

Sprint 4 (Day 10 – Day 12):

Goal is to test backend and system integration

Tasks:

- Write unit tests using Pytest (TDD)
- Test database layer
- Test authentication module
- Test detector module
- Test camera streaming module
- Integrate backend with frontend dashboard

MVP:

Reliable backend with tested modules and successful integration between frontend, backend, database, and AI model.

Sprint 5 (Day 13 – Day 15):

Goal is to optimize the project, complete the documentation and deploy the project

Tasks:

- Optimize backend performance
- Improve error handling and logging
- Final bug fixing
- Prepare UML diagrams and API documentation

- Perform end-to-end system testing
- Deploy and demonstrate complete application
- Optimize backend performance
- Improve error handling and logging
- Final bug fixing
- Prepare UML diagrams and API documentation
- Perform end-to-end system testing
- Deploy and demonstrate complete application

MVP:

A production-ready PPE Compliance Detection System capable of monitoring live CCTV feeds, detecting PPE violations in real time, storing events, generating alerts, and providing a dashboard for monitoring.